

Example of LLM for chemical planning

Prompt: You are given three different solutions (A, B, and C), each containing a different pair of inorganic salts from the following list, but you do not know which pair of salts are present in each solution: KCl , AgNO_3 , $\text{Pb}(\text{NO}_3)_2$, BaI_2 , Na_2SO_4 , $\text{Ni}(\text{ClO}_4)_2$ in addition to the three unknown solutions, you have access to 1M KOH and 1M HNO_3 and no other solution. You can mix the unknown solutions together or add acid or base to them. Your task is to identify which pair of salts is present in each solution. Decompose this problem into a sequence of experiments. Explain which tests to perform, how to measure outcomes, and how to iteratively refine the hypotheses to reach the final conclusion.

Answer:

Step 1: Analyze tools and constrains we have observation types:

Solubility rules:

- Ag^+ : insoluble with Cl^- (AgCl ppt), soluble otherwise
- Ba^{2+} : insoluble with SO_4^{2-} (BaSO_4 ppt)



Incomplete domain knowledge:
Omits Ag_2SO_4 , AgI precipitation

Step 2: Individual base test (add KOH)

Identify Ag^+ (brown/black), Pb^{2+} (white), Ni^{2+} (green) by ppt color

Step 3: Identify halides and sulfates via mixing

Mix unknown solutions in pairs (A+B, A+C, B+C). Observe halide/sulfate precipitation:

AgCl (white ppt), PbI_2 (yellow ppt), PbSO_4 (white ppt), BaSO_4 (ppt).

Step 4: Dissolve ppts with HNO_3

Ag_2O , $\text{Pb}(\text{OH})_2$, $\text{Ni}(\text{OH})_2$ dissolve in HNO_3

- Take ppt formed in step 2
- If ppt dissolves: confirms the cation identify



Redundant step: Unnecessary confirmation steps with no new information

Step 5: Iteratively refine assignment

Suppose after KOH addition:

A $\rightarrow$ green ppt $\rightarrow$ contains Ni^{2+}

B $\rightarrow$ brown ppt $\rightarrow$ contains Ag^+

.....

Then mix solutions:

B + A $\rightarrow$ white ppt $\rightarrow$ indicates Ag^+ + Cl^-

$\rightarrow$ solution B has Ag^+ , A has Cl^-

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Combine observation:

A $\rightarrow$ Ni^{2+} + Cl^- ?

B $\rightarrow$ Ag^+ + I^- ?

C $\rightarrow$ Pb^{2+} + SO_4^{2-} ?



Systematic incompleteness:
Only partial ions are recognized, not two complete salts in one solution



Chemical error: Ag^+ and I^- cannot coexist in same solution (B), Pb^{2+} and SO_4^{2-} cannot coexist in same solution (C)

Step 6: Cross-check final assignments

Mix assigned solutions $\rightarrow$ no new ppt unless expected.

Dissolve expected hydroxides with HNO_3 $\rightarrow$ confirms cation identity.

Double-check unusual cations like K^+ and Na^+ (remain in solution, soluble).